HRMS

Human Resource Management System

Functional Design Document

All Modules | User Flows | Business Rules | Screen Descriptions | Function Signatures

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Modules Covered	15 Modules — Complete System

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1. Document Purpose & Scope

This Functional Design Document (FDD) defines the detailed functional behaviour of all 15 modules of the HRMS custom ERP application. It serves as the primary reference for UI/UX designers, developers, and QA engineers during implementation.

For each module, this document provides: the functional overview, all user roles involved, step-by-step user flows, screen-level field and action descriptions, all business rules and validations, and the key function signatures with their inputs, logic summary, and outputs.

> **All business rules in this document are enforced at runtime using configuration entities. No rule is hardcoded. Refer to the Solution Design Document (SDD v2.0) for the full Configuration Entity Catalog.**

2. System Actors Overview

Actor	System Role	Primary Modules
Employee	End user — manages own profile, tasks, assets	Employee Master, Task Reporting, Time Reporting, Asset Registry, Beneficiary Mgmt
Manager	Team lead — assigns work, reviews delivery, rates quality	Task Reporting, Time Reporting, Efficiency Analytics, Asset Registry
HR Admin	Configures all HR rules, manages all employee data	All modules — full access to configuration and employee data
Finance Admin	Manages salary structure, reward formulas, payouts	Salary Management, Reward & Incentive, Beneficiary Mgmt, Reports
System Admin	Manages users, roles, system config, audit logs	Admin Dashboard, Role & Permission Config, Audit Logs, Notification Config

3. Employee Master

3.1 Overview

The Employee Master is the central profile record for every employee. All other modules reference this record. It stores personal, employment, professional, and skill data and automatically triggers BPV recalculation when qualifying fields are updated.

Actors / Roles	Description
Employee	Creates and maintains own profile information
HR Admin	Creates new employees, manages employment details, processes exits
Manager	Views team member profiles; cannot edit
System Admin	Views all records; manages user account linked to profile

3.2 User Flows & Workflows

Flow: Create New Employee

1. HR Admin opens Employee Master → New Employee.
2. Enters mandatory personal details: full name, date of birth, gender, contact email, phone.
3. Enters employment details: designation (from Designation Master), department (from Department Master), reporting manager, date of joining.
4. System auto-generates unique Employee ID.
5. HR Admin saves record. System creates employee profile with status Active.
6. System triggers BPV Engine to perform initial BPV calculation.
7. System creates linked user account and sends onboarding notification to employee.

Flow: Employee Updates Own Profile

1. Employee logs in and navigates to My Profile.
2. Employee selects the tab to update: Qualifications, Experience, Skills, or Personal.
3. Employee enters new data and saves.
4. System validates against relevant Master (Qualification Master, Skill Master, etc.).
5. On save, system queues a BPV recalculation job.
6. Employee receives notification: 'Your profile has been updated. BPV score will be recalculated.'
7. BPV Engine runs recalculation and updates the BPV Score on the profile.

Flow: Employee Exit Process

1. HR Admin opens the employee record and selects 'Process Exit'.
2. HR Admin enters exit date, exit reason, and notice period details.
3. System changes employee status to 'On Notice'.

4. System generates an Exit Handover Checklist listing all assets owned by the employee.
5. HR Admin assigns handover recipients for each asset.
6. On exit date, system changes status to 'Exited'. Operational ownership of all assets transfers to the organization.
7. Historical authorship and BPV history are permanently retained. Employee record is archived, not deleted.

3.3 Screen Descriptions

Screen: Employee List Screen

Fields / Sections:

- Search bar: search by name, ID, department, designation, status
- Filter panel: department, designation, status, date of joining range
- Data table: Employee ID | Name | Designation | Department | Manager | BPV Score | Status
- Pagination and export to CSV

Actions:

- ◦ Click row → open Employee Profile
- ◦ New Employee button → Create Employee form
- ◦ Bulk export selected rows

Screen: Employee Profile Screen — Personal Tab

Fields / Sections:

- Full Name (text, mandatory)
- Date of Birth (date picker, mandatory)
- Gender (dropdown from config)
- Contact Email (email, mandatory, unique)
- Phone Number (text)
- Emergency Contact Name and Phone
- Profile Photo (image upload, optional)

Actions:

- ◦ Edit → inline edit mode
- ◦ Save Changes
- ◦ Cancel

Screen: Employee Profile Screen — Employment Tab

Fields / Sections:

- Employee ID (auto-generated, read-only)
- Designation (dropdown from Designation Master)

- Department (dropdown from Department Master)
- Reporting Manager (employee lookup)
- Date of Joining (date picker)
- Employment Status (Active / On Notice / Exited, read-only — system managed)
- Contract Type (from config)

Actions:

- Edit (HR Admin only)
- Process Exit (HR Admin only)

Screen: Employee Profile Screen — Qualifications Tab

Fields / Sections:

- Qualification Type (dropdown from Qualification Master)
- Institution Name (text)
- Year of Completion (year picker)
- Grade / Score (optional text)
- Certificate Upload (file, optional)
- Add Multiple Qualifications supported

Actions:

- Add Qualification
- Edit Qualification
- Remove Qualification (with confirmation)

Screen: Employee Profile Screen — Skills Tab

Fields / Sections:

- Skill Category (from Skill Category Master)
- Skill Name (from Skill Master, filtered by category)
- Proficiency Level (from Skill Proficiency Master)
- Years of Experience with Skill (number)
- Add Multiple Skills supported

Actions:

- Add Skill
- Update Proficiency
- Remove Skill

Screen: Employee Profile Screen — BPV Tab

Fields / Sections:

- Current BPV Score (large numeric display, read-only)

- Score Breakdown: Education | Experience | Org Profile | Skills | Certifications (each as a sub-score)
- Last Calculated Date
- BPV History table: date | score | triggered by
- Salary Component 1 linked to current BPV Band (read-only for employee)

Actions:

  - Recalculate BPV (HR Admin only)
  - View Full BPV History

3.4 Business Rules & Validations

- **BR-01:** Employee ID is auto-generated by the system — format configurable by System Admin. It cannot be manually entered or edited.
- **BR-02:** Contact email must be unique across all employee records including exited employees.
- **BR-03:** Mandatory fields: Full Name, Date of Birth, Email, Designation, Department, Date of Joining. Form cannot be submitted if any mandatory field is empty.
- **BR-04:** Designation and Department must resolve to active entries in their respective Master tables.
- **BR-05:** On any change to Qualifications, Skills, Certifications, or Experience, BPV recalculation is automatically queued.
- **BR-06:** Employee can only edit personal and professional profile fields. Employment details (designation, department, manager) can only be edited by HR Admin.
- **BR-07:** Employee status transitions: Active → On Notice → Exited. Transitions cannot be reversed.
- **BR-08:** On exit, the system must verify that all owned assets have been assigned a handover recipient before the exit can be confirmed.
- **BR-09:** Exited employee records are archived. They cannot be deleted. Historical data must be retained indefinitely.
- **BR-10:** A manager cannot be assigned as their own reporting manager.

3.5 Function Signatures

Function	Input	Process Summary	Output
createEmployee(data)	EmployeeData object	Validates data, generates Employee ID, creates profile, triggers BPV calculation, sends onboarding notification	EmployeeProfile | ValidationError
updateEmployeeProfile(empId, data)	Employee ID, updated fields	Validates fields against Masters, saves changes, queues BPV recalculation if qualifying fields changed	UpdatedProfile | ValidationError
processEmployeeExit(empId,	Employee ID, exit	Validates handover	ExitConfirmation |

Function	Input	Process Summary	Output
exitData)	date, reason	completion, updates status, transfers asset ownership, archives profile	HandoverPendingError
getEmployeeProfile(empId)	Employee ID	Fetches profile with all tabs and linked BPV data	EmployeeProfile
searchEmployees(filters)	Filter criteria object	Queries employee records with filters, returns paginated list	PaginatedEmployeeList

4. Base Professional Value (BPV) Engine

4.1 Overview

The BPV Engine calculates each employee's Base Professional Value score using configuration-driven weighted scoring. It is triggered automatically on profile changes and on scheduled review cycles. No scoring formula or weight is hardcoded — all values are read from the active configuration at runtime.

Actors / Roles	Description
System (Automated)	Triggers BPV calculation on profile change or review cycle
HR Admin	Configures BPV weights, triggers manual recalculation, views history
Employee	Views own BPV score and breakdown — read only
Manager	Views team BPV scores — read only

4.2 User Flows & Workflows

Flow: Automatic BPV Recalculation on Profile Change

1. Employee or HR Admin updates a qualifying field (qualification, experience, skill, certification).
2. System detects the change and queues a BPV recalculation job.
3. BPV Engine reads active BPV Weight Config (effective_to is null).
4. Engine calculates Education Score from Qualification Master.
5. Engine calculates Experience Score from Experience Band Master and Experience Type Master.
6. Engine calculates Organization Score from Organization Type Master.
7. Engine calculates Skill Score from Skill Master and Skill Proficiency Master.
8. Engine calculates Certification Score from Certification Master.
9. Final BPV = weighted sum of all factor scores.
10. Score is capped if BPV Score Cap Config is set for the employee's grade.
11. New BPV record is inserted in BPV history table with timestamp, score, config version used, and trigger reason.
12. Employee profile BPV Score field is updated.
13. Salary Component 1 is flagged for recalculation at the next review cycle.

Flow: Scheduled Bulk BPV Recalculation

1. System runs bulk recalculation job per Review Cycle Config (monthly / quarterly / annually).
2. Job iterates over all active employees.
3. For each employee, calculateBPV() is called.
4. On completion, Salary Component 1 is updated per Salary Band Master.
5. HR Admin receives a bulk recalculation completion report.

Flow: HR Admin Manual Recalculation

1. HR Admin opens employee BPV tab and clicks Recalculate BPV.
2. System prompts for a mandatory reason.
3. BPV Engine runs calculation, stores result with reason and HR Admin's name.
4. HR Admin views updated score and breakdown.

Flow: Configure BPV Weights

1. HR Admin opens Admin Dashboard → BPV Weight Config.
2. HR Admin views current active weights (must sum to 100%).
3. HR Admin adjusts weights and enters effective date.
4. System validates that new weights sum to 100%.
5. System saves new config version. Old config is closed with effective_to = day before new effective_from.
6. HR Admin receives confirmation. New weights apply to all future calculations.

4.3 Screen Descriptions

Screen: BPV Score Dashboard (HR Admin View)

Fields / Sections:

- Employee search and filter panel
- Data table: Employee Name | Designation | Dept | BPV Score | Last Calculated | Salary Band | Score Trend (↑ ↓)
- Summary: avg BPV by department, top 10 scorers

Actions:

  - View individual BPV details
  - Trigger bulk recalculation
  - Export to Excel

Screen: BPV Score Detail Screen

Fields / Sections:

- Employee name and ID (header)
- Overall BPV Score (prominent display)
- Factor breakdown bar chart: Education | Experience | Org | Skills | Certifications
- Each factor: raw score, weight applied, weighted score
- Config version used for calculation (with effective date)
- Calculation timestamp and trigger reason

Actions:

  - View BPV History
  - Trigger Manual Recalculation (HR Admin only)

Screen: BPV History Screen

Fields / Sections:

- Chronological table: date | BPV score | trigger (auto / scheduled / manual) | triggered by | config version | change vs previous
- Filter by date range
- Delta chart: BPV score over time

Actions:

  - View snapshot of config used at a specific calculation
  - Export history

Screen: BPV Weight Configuration Screen (Admin)

Fields / Sections:

- Current active config: each factor name | current weight % | effective from date
- New config entry form: weights for all 5 factors | effective date
- Validation indicator: weights sum indicator (must reach 100%)
- Version history table

Actions:

  - Save New Config Version
  - View Old Config Versions
  - Rollback View (read-only)

4.4 Business Rules & Validations

- **BR-01:** BPV calculation must use the configuration version that was active at the calculation timestamp — not the current latest version.
- **BR-02:** BPV weights (Education, Experience, Org, Skills, Certifications) must sum to exactly 100%. System rejects config save if they do not.
- **BR-03:** Every BPV calculation stores a snapshot of: the config version used, all input factor scores, final score, and trigger reason. This snapshot is immutable.
- **BR-04:** If no Qualification Master entry exists for a qualification held by the employee, that qualification contributes zero score and is flagged for HR review.
- **BR-05:** BPV Score Cap Config is applied after all factor scores are summed. If the total exceeds the cap, the score is set to the cap value. Cap overrides are logged.
- **BR-06:** Scheduled recalculation cannot overwrite a manual recalculation performed on the same day. Manual takes precedence.
- **BR-07:** BPV history records are append-only. No BPV record can be edited or deleted.
- **BR-08:** HR Admin must provide a reason (minimum 10 characters) for any manual BPV recalculation.

4.5 Function Signatures

Function	Input	Process Summary	Output
calculateBPV(empId, configVersion?)	Employee ID, optional config version override	Reads active config, computes all factor scores, applies weights, applies cap, stores history record	BPVResult {score, breakdown, configVersion, timestamp}
getBPVHistory(empId, dateRange?)	Employee ID, optional date range	Returns all BPV history records for the employee	BPVHistory[]
getFactorScore(empId, factor)	Employee ID, factor name	Computes score for one factor (education/experience/org/skills/certs) using active config	FactorScore {raw, weight, weighted}
bulkRecalculateBPV(filters?)	Optional department/grade filter	Queues BPV recalculation for all matching active employees, updates salary bands	BulkJobResult {processed, failed, reportUrl}
saveBPVWeightConfig(weights, effectiveDate)	Weight object {edu, exp, org, skill, cert}, effective date	Validates sum=100%, closes current config, creates new versioned record	ConfigVersion | ValidationError

5. Qualification & Experience Scoring

5.1 Overview

This module manages the entry and scoring of employee academic qualifications, professional experience, and organization history. It feeds directly into the BPV Engine as the Education Score, Experience Score, and Organization Score components.

Actors / Roles	Description
Employee	Enters and updates own qualifications, experience, and org history
HR Admin	Reviews, verifies, and manages all employee qualification and experience data; manages Qualification Master and Experience Band Master

5.2 User Flows & Workflows

Flow: Add Qualification

1. Employee navigates to Profile → Qualifications tab.
2. Clicks Add Qualification.
3. Selects Qualification Type from dropdown (from Qualification Master).
4. Enters institution name, year of completion, and optional grade.
5. Uploads certificate if available (PDF/image, max 5MB).
6. Saves. System records qualification and triggers BPV recalculation.

Flow: Add Professional Experience Entry

1. Employee navigates to Profile → Experience tab.
2. Clicks Add Experience.
3. Enters: organization name, organization type (from Organization Type Master), start date, end date (or 'Current'), designation held, domain, experience type (domain / enterprise / global / leadership).
4. Saves. System calculates years for this entry automatically from start/end dates.
5. BPV recalculation is triggered.

Flow: HR Admin Verify Qualification

1. HR Admin opens employee profile → Qualifications tab.
2. Reviews submitted qualifications and uploaded certificates.
3. Marks each as Verified or Flagged (with a reason note).
4. Flagged qualifications are highlighted on the employee's profile and contribute zero to BPV until verified.

5.3 Screen Descriptions

Screen: Qualifications Tab

Fields / Sections:

- Table of existing qualifications: Type | Institution | Year | Grade | Certificate | Verification Status
- Add Qualification button
- Each row: inline Edit and Remove actions

Actions:

  - Add Qualification
  - Edit Qualification
  - Remove (with confirmation)
  - Verify / Flag (HR Admin only)

Screen: Experience History Tab

Fields / Sections:

- Timeline view of experience entries: org name | org type | designation | start–end date | years calculated | experience types
- Total years of experience (auto-summed, displayed at top)
- Add Experience button

Actions:

  - Add Experience
  - Edit Experience Entry
  - Remove Entry
  - Mark as Current Position

5.4 Business Rules & Validations

- **BR-01:** Qualification type must be selected from Qualification Master. Free-text qualification types are not permitted.
- **BR-02:** An employee can hold multiple qualifications. All are stored and scored. The system does not restrict to a single 'highest' qualification — all contribute cumulatively as configured.
- **BR-03:** Year of completion cannot be in the future.
- **BR-04:** Experience end date cannot be before start date.
- **BR-05:** Only one experience entry can be marked as 'Current'. Marking a new entry as current auto-sets the previous current entry's end date.
- **BR-06:** Organization Type must be selected from Organization Type Master. Free-text org types are not permitted.
- **BR-07:** Flagged (unverified) qualifications contribute zero score to BPV until an HR Admin marks them Verified.
- **BR-08:** Experience years for each entry are calculated as: (end date - start date) in decimal years, rounded to 2 decimal places. For 'Current' entries, today's date is used as end date.

- **BR-09:** Total experience years = sum of years across all non-overlapping experience entries. Overlapping periods are deduplicated.
- **BR-10:** Certificate uploads are stored securely and linked to the qualification record. File size limit and accepted formats are configurable.

5.5 Function Signatures

Function	Input	Process Summary	Output
addQualification(empId, data)	Employee ID, qualification data	Validates against Qualification Master, saves record, queues BPV recalculation	QualificationRecord | ValidationError
addExperienceEntry(empId, data)	Employee ID, experience data	Validates dates and org type, calculates years, saves entry, queues BPV recalculation	ExperienceEntry | ValidationError
verifyQualification(qualId, status, note)	Qualification ID, status (verified/flagged), optional note	Updates verification status, triggers BPV recalculation	UpdatedQualification
getEducationScore(empId)	Employee ID	Sums scores for all verified qualifications from Qualification Master	EducationScore {total, breakdown[]}
getExperienceScore(empId)	Employee ID	Calculates score from experience bands and types using Experience Band Master and Experience Type Master	ExperienceScore {total, byType{}}

6. Technical Skill & Certification Management

6.1 Overview

This module manages the employee's technical skills, proficiency levels, and professional certifications. It feeds into the BPV Engine as the Skill Score and Certification Score. Certification expiry is tracked and triggers alerts automatically.

Actors / Roles	Description
Employee	Adds and updates skills and certifications
HR Admin	Manages Skill Master, Certification Master, proficiency levels; reviews skill profiles
Manager	Views team skill matrix for resource planning (read-only)

6.2 User Flows & Workflows

Flow: Add Technical Skill

1. Employee navigates to Profile → Skills tab.
2. Clicks Add Skill.
3. Selects Skill Category (from Skill Category Master), then Skill Name (from Skill Master filtered by category).
4. Selects Proficiency Level (from Skill Proficiency Master: Beginner / Intermediate / Expert or as configured).
5. Enters years of experience with this skill.
6. Saves. System triggers BPV recalculation.

Flow: Add Professional Certification

1. Employee navigates to Profile → Certifications tab.
2. Clicks Add Certification.
3. Selects Certification from Certification Master (or requests HR to add a new certification).
4. Enters issue date and expiry date.
5. Uploads certificate document.
6. Saves. System triggers BPV recalculation and schedules expiry alert.

Flow: Certification Expiry Alert

1. System runs a daily job checking certifications expiring within the alert window (configurable, e.g., 60 days).
2. Employee receives notification: 'Your [Certification Name] expires on [date]. Please renew.'
3. HR Admin also receives a summary report of expiring certifications.
4. On expiry date, certification status changes to Expired. Its BPV score contribution drops to zero.

5. Employee can upload renewed certification — a new record is created, old record is archived.

Flow: Team Skill Matrix View (Manager)

1. Manager navigates to Team → Skill Matrix.
2. System displays a matrix: employees (rows) vs. skills (columns), with proficiency levels filled in.
3. Manager can filter by skill category, proficiency level, or individual skill.
4. Manager can identify skill gaps across the team.

6.3 Screen Descriptions

Screen: Skills Tab

Fields / Sections:

- Grid of current skills: Category | Skill | Proficiency | Years | BPV Score Contribution
- Add Skill button
- Search/filter by category

Actions:

  - Add Skill
  - Edit Proficiency / Years
  - Remove Skill (with confirmation)

Screen: Certifications Tab

Fields / Sections:

- Table: Certification Name | Issuing Body | Issue Date | Expiry Date | Status (Active/Expiring Soon/Expired) | BPV Contribution
- Color-coded status: green=active, amber=expiring, red=expired
- Add Certification button

Actions:

  - Add Certification
  - View Certificate
  - Renew (upload new)
  - Remove

Screen: Team Skill Matrix Screen

Fields / Sections:

- Column headers: Skill names (filterable)
- Row headers: Team member names
- Cells: Proficiency level (B/I/E) or empty if not held
- Summary row: team coverage % per skill

Actions:

- Filter by category
- Export matrix to Excel
- Click cell to view full employee profile

6.4 Business Rules & Validations

- **BR-01:** Skill must be selected from Skill Master under a valid Skill Category. Custom free-text skills are not allowed.
- **BR-02:** Proficiency level must be selected from Skill Proficiency Master. The number of proficiency levels is configurable.
- **BR-03:** An employee can hold the same skill only once — duplicate skill entries are not permitted. Proficiency update replaces the existing record.
- **BR-04:** Certification must be selected from Certification Master. If a certification is not listed, the employee requests HR Admin to add it to the Master.
- **BR-05:** Expiry date is mandatory for certifications that have an expiry. For lifetime certifications, this field is marked 'No Expiry' (configurable per certification in Certification Master).
- **BR-06:** Expired certifications contribute zero to BPV Score. The BPV Engine checks certification status (active vs. expired) at calculation time.
- **BR-07:** Certification expiry alert window is configurable in System Config (default: 60 days before expiry).
- **BR-08:** When a certification is renewed, the old record is archived (not deleted) and a new active record is created. BPV recalculation is triggered automatically.

6.5 Function Signatures

Function	Input	Process Summary	Output
addSkill(empId, skillId, proficiencyId, years)	Employee ID, Skill ID, Proficiency Level ID, years of experience	Validates uniqueness, saves skill record, triggers BPV recalculation	SkillRecord | DuplicateError
updateSkillProficiency(empSkillId, proficiencyId)	Employee-Skill record ID, new proficiency	Updates proficiency, triggers BPV recalculation	UpdatedSkillRecord
addCertification(empId, certId, issueDate, expiryDate, fileRef)	Employee ID, Certification ID, dates, uploaded file reference	Saves certification, schedules expiry alert, triggers BPV recalculation	CertificationRecord | ValidationError
checkExpiringCertifications()	None (system job)	Queries all certifications expiring within alert window, sends notifications, flags expired	AlertsSent, StatusesUpdated
getSkillMatrix(managerId)	Manager ID	Returns team skill matrix: employees × skills × proficiency levels	SkillMatrix[][]

7. Outcome-Based Task Reporting

7.1 Overview

This module replaces traditional timesheet tracking with an outcome-based model. Managers assign tasks with deliverables and quality criteria. Employees report completion. Managers review and rate quality. Task outcomes feed directly into Efficiency Scoring.

Actors / Roles	Description
Manager	Assigns tasks, reviews delivery, rates quality, manages task status
Employee	Views assigned tasks, submits task completion with quality self-assessment
HR Admin	Views all task data, configures task categories and quality criteria

7.2 User Flows & Workflows

Flow: Manager Assigns Task

1. Manager navigates to Team → Tasks → New Task.
2. Enters task title, description, and detailed deliverables.
3. Selects task category (from Task Category Master — determines benchmark duration).
4. Assigns to employee (from team member list).
5. Sets expected completion date.
6. System automatically attaches the quality criteria checklist for the selected category (from Quality Criteria Master).
7. Saves. Employee receives a task assignment notification.

Flow: Employee Submits Task Completion

1. Employee navigates to My Tasks → selects the task.
2. Clicks Submit Completion.
3. Selects completion status: Fully Completed / Partially Completed / Blocked.
4. For each quality criteria item in the checklist, marks self-assessment: Met / Not Met / N/A.
5. Enters any blocker or dependency notes (mandatory if status is Blocked).
6. Saves. Task status changes to 'Pending Review'. Manager is notified.

Flow: Manager Reviews and Rates Task

1. Manager navigates to Team → Tasks → Pending Review.
2. Opens the task. Reviews submitted deliverables and employee self-assessment.
3. Rates each quality criteria item: Met / Partially Met / Not Met.
4. Selects overall outcome: Accepted / Returned for Revision.

5. If Returned: enters mandatory revision notes. Task status reverts to In Progress.
6. If Accepted: enters quality rating from Quality Rating Master scale.
7. System records quality rating and marks task as Accepted.
8. Accepted task becomes eligible for time logging.

7.3 Screen Descriptions

Screen: Task List Screen — Employee View

Fields / Sections:

- Filter tabs: All | In Progress | Pending Review | Accepted | Returned
- Task card: title | assigned by | task category | expected date | status badge | benchmark duration (read-only)
- Search by task title

Actions:

  - Open Task Detail
  - Submit Completion

Screen: Task Detail Screen — Employee View

Fields / Sections:

- Task Title and Description (read-only)
- Deliverables list (read-only)
- Quality Criteria Checklist (self-assessment — Met / Not Met / N/A per item)
- Completion Status selector
- Blocker Notes text area (conditionally mandatory)
- Submission timestamp on submit

Actions:

  - Submit Completion
  - Save Draft

Screen: Task Review Screen — Manager View

Fields / Sections:

- Employee self-assessment display (read-only)
- Quality Criteria: each item with employee response + manager rating input
- Overall Quality Rating dropdown (from Quality Rating Master)
- Outcome selector: Accepted / Returned for Revision
- Revision Notes (mandatory if Returned)

Actions:

  - Accept Task

- Return for Revision
- View Task History (all submissions and reviews)

Screen: Team Task Dashboard — Manager View

Fields / Sections:

- Summary cards: Total Assigned | Pending Review | Overdue | Accepted this period
- Filter: by employee, category, date range, status
- Table: task title | assignee | category | assigned date | expected date | status | quality rating

Actions:

- Open Task
- New Task
- Export

7.4 Business Rules & Validations

- **BR-01:** A task must be assigned to exactly one employee. Group tasks are not supported in this version.
- **BR-02:** Task Category is mandatory — it determines the benchmark duration used in Efficiency Scoring and attaches the correct quality criteria checklist.
- **BR-03:** An employee can only submit completion for tasks assigned to them.
- **BR-04:** Completion status 'Blocked' requires a mandatory blocker notes entry.
- **BR-05:** A task can be returned for revision a maximum number of times (configurable in system config, default: 3). Beyond this limit, the task is escalated to HR Admin.
- **BR-06:** Quality rating is mandatory when a manager accepts a task. It must be selected from the active Quality Rating Master scale.
- **BR-07:** A task must be Accepted before the employee can log time against it.
- **BR-08:** Overdue detection: a task is marked Overdue if its expected completion date has passed and its status is not Accepted. A daily job checks and flags overdue tasks.
- **BR-09:** Task records are immutable once Accepted — the final quality rating and acceptance timestamp cannot be changed.
- **BR-10:** All task submissions and review actions are timestamped and attributed to the acting user.

7.5 Function Signatures

Function	Input	Process Summary	Output
createTask(managerId, taskData)	Manager ID, task data (title, description, deliverables, category, assignee, due date)	Validates category and assignee, attaches quality criteria, creates task, notifies employee	Task | ValidationError
submitTaskCompletion(taskId,	Task ID, Employee ID,	Validates ownership, checks	UpdatedTask |

Function	Input	Process Summary	Output
empId, completionData)	completion status, self-assessment, blocker notes	mandatory fields, updates task status, notifies manager	PermissionError
reviewTask(taskId, managerId, reviewData)	Task ID, Manager ID, quality ratings, outcome, revision notes	Validates outcome and rating, updates task status, triggers revision or acceptance flow	ReviewedTask | ValidationError
getMyTasks(empId, filters)	Employee ID, optional filters	Returns paginated task list for the employee with status and metadata	PaginatedTaskList
getTeamTasks(managerId, filters)	Manager ID, optional filters	Returns all tasks assigned by manager, with status summary	TeamTaskList
getOverdueTasks()	None (system job)	Flags tasks past due date with non-accepted status, notifies manager	OverdueTaskList

8. Time Reporting

8.1 Overview

Time Reporting is a minimal operational record — it does not measure performance, only captures actual time consumed for reference and Component 2 salary calculation. Time entries can only be submitted against Accepted tasks. Allowed time blocks are fully configurable.

Actors / Roles	Description
Employee	Submits time entries against accepted tasks
Manager	Reviews and approves team time entries
Finance Admin	Views time data for Component 2 salary calculation
HR Admin	Configures Time Block Master

8.2 User Flows & Workflows

Flow: Employee Logs Time

1. Employee navigates to Time Reporting → New Entry.
2. Selects the date (cannot be future date).
3. Selects the task (only Accepted tasks appear in the dropdown).
4. Selects time block (from Time Block Master: e.g., 1 hr / 2 hrs / 4 hrs / 6 hrs).
5. Optionally enters a note.
6. Submits. Status: Pending Approval.
7. Manager receives notification of new time entry.

Flow: Manager Approves Time Entry

1. Manager navigates to Team → Time Entries → Pending Approval.
2. Reviews the time entry: employee, date, task, time block.
3. Approves or Rejects with a mandatory reason if rejected.
4. Approved entries are locked. Rejected entries are returned to employee for correction.

Flow: Weekly Time Summary View

1. Employee views a weekly calendar-style grid of logged time.
2. Each day shows: task name, time block logged, status (pending/approved/rejected).
3. Total approved hours shown for the week.
4. Export available as PDF or CSV.

8.3 Screen Descriptions

Screen: Time Entry Form

Fields / Sections:

- Date picker (today or past dates only — future dates disabled)
- Task selector (dropdown — Accepted tasks only, with task name and category shown)
- Time Block selector (radio buttons or dropdown from Time Block Master)
- Notes (optional text area, max 200 characters)

Actions:

  - Submit Entry
  - Save Draft

Screen: Time Entry List — Employee View

Fields / Sections:

- Calendar or table view: date | task | time block | status | approved by
- Filter: by date range, task, status
- Weekly and monthly totals

Actions:

  - Submit New Entry
  - Edit Draft Entry
  - View Approved Entry (read-only)

Screen: Team Time Approval Screen — Manager View

Fields / Sections:

- Filter: by employee, date range, status
- Table: employee name | date | task | time block | submission timestamp | status
- Bulk approve option

Actions:

  - Approve
  - Reject with reason
  - View Task Details

8.4 Business Rules & Validations

- **BR-01:** Time entries can only be submitted against tasks with status Accepted. Pending, In Progress, and Returned tasks are not eligible.
- **BR-02:** Date must not be in the future. System enforces this at submission time.
- **BR-03:** Time block must be selected from the active Time Block Master. Free-text time values are not permitted.
- **BR-04:** Only one time entry is permitted per employee per task per date. A duplicate submission is rejected.

- **BR-05:** Time entries require manager approval before they are included in Component 2 salary calculation.
- **BR-06:** Approved time entries are locked — neither the employee nor the manager can edit them. Only Finance Admin can make corrections with a mandatory audit log entry.
- **BR-07:** Rejected time entries must include a rejection reason (minimum 10 characters). The employee is notified and can resubmit with corrections.
- **BR-08:** Component 2 salary is calculated from Approved time entries only — Pending and Rejected entries are excluded.
- **BR-09:** The salary calculation period aligns with the Review Cycle Config (monthly by default).

8.5 Function Signatures

Function	Input	Process Summary	Output
createTimeEntry(empId, data)	Employee ID, date, taskId, timeBlockId, note	Validates date, task status, duplicate check, saves entry with Pending status, notifies manager	TimeEntry | ValidationError
reviewTimeEntry(entryId, managerId, decision, reason?)	Entry ID, Manager ID, Approved/Rejected, optional reason	Updates status, locks if approved, notifies employee if rejected	UpdatedTimeEntry
getMyTimeEntries(empId, dateRange)	Employee ID, date range	Returns all time entries for the period with totals	TimeEntrySummary
getComponent2Value(empId, period)	Employee ID, salary period	Sums approved time entries for the period, applies rate from Salary Grade Master, returns Component 2 amount	Component2Amount
getTeamTimeReport(managerId, period)	Manager ID, reporting period	Returns all team time entries for the period with approval status	TeamTimeReport

9. Dual-Component Salary Management

9.1 Overview

The salary module calculates and manages the two salary components for each employee. Component 1 is derived from the BPV score via the Salary Band Master. Component 2 is computed from approved time entries. All split ratios, salary bands, and rates are metadata-driven.

Actors / Roles	Description
Finance Admin	Configures salary bands, split ratios, reviews salary calculations, approves revisions
HR Admin	Initiates salary reviews, views employee salary summaries
Employee	Views own salary breakdown — read only

9.2 User Flows & Workflows

Flow: Automatic Salary Calculation on Review Cycle

1. Review Cycle Config triggers salary calculation (monthly / quarterly / annually).
2. System reads Salary Grade Master for each employee's grade → fetches Component 1/2 split ratio.
3. System reads BPV Score → maps to salary range in Salary Band Master → derives Component 1 amount.
4. System calls getComponent2Value() to compute time-based Component 2 amount.
5. Total salary = Component 1 + Component 2.
6. Finance Admin receives calculation summary for review.
7. Finance Admin approves. Salary records are updated and locked for the period.

Flow: Finance Admin Reviews and Adjusts Salary Band Config

1. Finance Admin opens Admin Dashboard → Salary Band Master.
2. Views current BPV score ranges and associated salary ranges per grade.
3. Adjusts ranges or salary values.
4. Enters effective date.
5. System versions the change. Old bands retained for historical calculations.

Flow: Employee Views Salary Breakdown

1. Employee opens My Profile → Salary tab.
2. Views: Component 1 amount, Component 2 amount, total salary, period.
3. Can drill into BPV score that drove Component 1.
4. Can view approved time entries that drove Component 2.
5. Historical salary records by period are viewable.

9.3 Screen Descriptions

Screen: Employee Salary View Tab

Fields / Sections:

- Current Period: Component 1 amount | Component 2 amount | Total
- BPV Score used → Salary Band applied → Component 1 derived (breakdown visible)
- Component 2: Approved hours this period × rate
- Salary history table: period | C1 | C2 | Total | Approved by

Actions:

  - View BPV Detail
  - View Time Entries for this period

Screen: Salary Grade Configuration Screen (Finance Admin)

Fields / Sections:

- Table: Grade | C1 Split % | C2 Split % | Component 2 Rate per time block
- Effective date
- Add/Edit Grade rows

Actions:

  - Save New Config Version
  - View Version History

Screen: Salary Band Configuration Screen (Finance Admin)

Fields / Sections:

- Table: Grade | BPV Score From | BPV Score To | Salary Range From | Salary Range To | Currency
- Rows are per grade per BPV band
- Add band row, edit existing, remove (soft delete with versioning)

Actions:

  - Save New Config Version
  - Preview: enter BPV score → see applicable band

9.4 Business Rules & Validations

- **BR-01:** Component 1 + Component 2 split percentages must sum to 100% for every grade in Salary Grade Master.
- **BR-02:** Component 1 amount is derived by mapping the employee's current BPV score to the active Salary Band Master. If the BPV score falls between two bands, the lower band applies (configurable: lower/upper/interpolation).

- **BR-03:** Component 2 is calculated only from Approved time entries within the salary period. Pending and Rejected entries are excluded.
- **BR-04:** Salary records for an approved period are locked. Corrections require Finance Admin override with a mandatory reason and audit log entry.
- **BR-05:** Currency is read from Currency Config. All salary amounts are stored in the configured base currency.
- **BR-06:** If an employee's BPV score exceeds the highest configured salary band for their grade, the system flags this for Finance Admin review — it does not auto-assign a salary above the defined bands.
- **BR-07:** Salary history is retained indefinitely. No salary record is deleted.
- **BR-08:** Salary data is accessible only to the employee (own data), HR Admin, and Finance Admin. Managers cannot view salary data.

9.5 Function Signatures

Function	Input	Process Summary	Output
calculateSalary(empId, period)	Employee ID, salary period	Reads BPV score, maps to salary band, computes C1; sums approved time entries for C2; returns total	SalaryCalculation {c1, c2, total, bpvUsed, bandApplied}
approveSalaryPeriod(period, financeAdminId)	Period, approver ID	Locks all salary records for the period, records approver and timestamp	ApprovalConfirmation
getSalaryHistory(empId)	Employee ID	Returns all salary records by period with component breakdown	SalaryHistory[]
saveSalaryBandConfig(grade, bands, effectiveDate)	Grade, array of BPV-to-salary band objects, effective date	Validates bands are contiguous and non-overlapping, creates versioned config	ConfigVersion | ValidationError
saveSalaryGradeConfig(gradeData, effectiveDate)	Grade split and rate data, effective date	Validates C1+C2 = 100%, creates versioned config	ConfigVersion | ValidationError

10. Productivity & Efficiency Analytics

10.1 Overview

The Efficiency Engine calculates a score for each employee based on three configurable factors: task completion rate, quality score, and time efficiency. This score maps to incentive eligibility via the Efficiency Tier Master. All weights and tier mappings are metadata-driven.

Actors / Roles	Description
System (Automated)	Calculates efficiency scores per the configured period
HR Admin	Views efficiency reports, configures weights and tiers
Manager	Views team efficiency scores and trends
Employee	Views own efficiency score and incentive eligibility
Finance Admin	Uses efficiency tier output for incentive calculation

10.2 User Flows & Workflows

Flow: Automated Efficiency Score Calculation

1. System runs efficiency calculation job per Efficiency Period Config.
2. For each employee, system computes three factor scores for the period:
3. a) Completion Rate = Accepted Tasks / Total Assigned Tasks × 100.
4. b) Quality Score = Average quality rating across all Accepted tasks for the period.
5. c) Time Efficiency = Benchmark Duration / Actual Reported Time × 100 (capped at 100 for tasks completed faster than benchmark).
6. System reads Efficiency Weight Config → applies weights to each factor.
7. Efficiency Score = (CR × CR weight) + (QS × QS weight) + (TE × TE weight).
8. Score is mapped to Efficiency Tier Master → incentive eligibility percentage is attached.
9. Efficiency record is stored with full calculation snapshot.

Flow: Employee Views Efficiency Score

1. Employee navigates to My Performance.
2. Views current period efficiency score, tier, and incentive eligibility %.
3. Views breakdown: completion rate, quality score, time efficiency — each with their weight applied.
4. Views trend chart of efficiency score over past periods.

Flow: Manager Views Team Efficiency

1. Manager navigates to Team → Efficiency.
2. Views team efficiency table: employee name | current score | tier | trend.
3. Clicks employee row to view full breakdown.
4. Can filter by period and sort by score.

10.3 Screen Descriptions

Screen: My Performance Screen — Employee

Fields / Sections:

- Efficiency Score (large display with color-coded tier: gold/silver/bronze)
- Incentive Eligibility % for the period
- Score breakdown: Completion Rate | Quality Score | Time Efficiency — each with weight applied
- Bar chart: score trend over last 6 periods
- Comparison to team average (anonymous)

Actions:

  - View Task Detail
  - Download Performance Report

Screen: Team Efficiency Dashboard — Manager

Fields / Sections:

- Table: Employee | Current Score | Tier | Completion Rate | Avg Quality | Time Efficiency | Trend ↑ ↓
- Period selector
- Team average scores row at the bottom

Actions:

  - View employee detail
  - Export
  - Filter by tier

Screen: Efficiency Configuration Screen — Admin

Fields / Sections:

- Weight Config: Completion Rate % | Quality Score % | Time Efficiency % (must sum to 100%)
- Tier Config table: score from | score to | tier name | incentive eligibility %
- Effective date for each config
- Version history

Actions:

  - Save New Config Version
  - Preview: enter scores → see tier

10.4 Business Rules & Validations

- **BR-01:** Efficiency weights (Completion Rate, Quality Score, Time Efficiency) must sum to exactly 100%. Config save is rejected if they do not.
- **BR-02:** Efficiency Score is calculated only if the employee has at least one Accepted task in the period. If no tasks were accepted, the score is marked as 'Insufficient Data' and the employee receives no incentive eligibility for the period.
- **BR-03:** Time Efficiency is capped at 100 — an employee who completes work faster than the benchmark does not score above 100 on that factor.
- **BR-04:** Quality Score is the average of all manager-assigned quality ratings for Accepted tasks in the period. It uses the numeric equivalent of the Quality Rating Master scale.
- **BR-05:** Efficiency records are append-only — one record per employee per period. Records cannot be edited after generation.
- **BR-06:** Incentive eligibility % from the Efficiency Tier Master is a reference for Finance. The actual incentive amount is calculated in the Reward & Incentive module.
- **BR-07:** If the Efficiency Tier Master has no tier covering the calculated score, the system flags this as a configuration gap and notifies HR Admin.

10.5 Function Signatures

Function	Input	Process Summary	Output
calculateEfficiencyScore(empId, period)	Employee ID, period	Computes CR, QS, TE; reads weights; calculates weighted score; maps to tier; stores snapshot	EfficiencyScore {score, tier, incentivePct, breakdown}
getEfficiencyReport(empId, periods?)	Employee ID, optional period range	Returns efficiency history with breakdown per period	EfficiencyHistory[]
getTeamEfficiency(managerId, period)	Manager ID, period	Returns all team members' efficiency scores for the period	TeamEfficiencyReport[]
saveEfficiencyWeightConfig(weights, effectiveDate)	Weight object, effective date	Validates sum=100%, creates versioned config	ConfigVersion | ValidationError
saveEfficiencyTierConfig(tiers, effectiveDate)	Array of tier objects, effective date	Validates tiers are contiguous and cover 0-100, creates versioned config	ConfigVersion | ValidationError

11. Data Ownership Registry

11.1 Overview

The Data Ownership Registry is a permanent, searchable record of every digital asset created or contributed to by employees. It tracks full ownership metadata, version history, and lifecycle status. Asset types are configurable. Authorship is immutable — it is retained permanently, even after employee exit.

Actors / Roles	Description
Employee	Registers owned assets, adds contributors, updates versions
Manager	Reviews team assets, approves new asset registrations
HR Admin	Manages Asset Type Master, views all assets, processes exit handovers
System	Auto-transfers ownership on employee exit, flags compliance gaps

11.2 User Flows & Workflows

Flow: Register New Asset

1. Employee navigates to Asset Registry → New Asset.
2. Enters asset name and description.
3. Selects asset type from Asset Type Master.
4. System attaches required documentation checklist from Doc Requirement Config.
5. Employee adds contributors (other employees who contributed).
6. Employee selects the reviewer and approver (from team/department).
7. Saves. Asset is created with status 'Pending Approval', version 1.0.
8. Reviewer is notified to review. Approver is notified to approve.
9. On approval: asset status changes to Active. Documentation compliance tracking begins.

Flow: Update Asset Version

1. Employee opens existing asset from My Assets.
2. Makes changes to asset details or uploads updated documentation.
3. Saves as new version — system auto-increments version number (e.g., 1.0 → 1.1 or 2.0).
4. Version type: minor (patch/update) or major (significant revision) — employee selects.
5. Previous version is archived. New version becomes active.
6. Version history is updated with: version number, date, changed by, change summary.

Flow: Exit Handover Process

1. HR Admin processes employee exit.

2. System generates Exit Handover Checklist: all assets where the exiting employee is Creator or Owner.
3. HR Admin assigns a handover recipient for each asset from the system.
4. On confirmation: Operational Owner field updated to handover recipient.
5. Historical Creator and Contributor fields remain unchanged — permanently attributed to the original employee.
6. Assets without a handover recipient block exit confirmation.

11.3 Screen Descriptions

Screen: Asset Registration Form

Fields / Sections:

- Asset Name (text, mandatory)
- Asset Type (dropdown from Asset Type Master, mandatory)
- Description (text area)
- Project / Department association
- Contributors (multi-select employee lookup)
- Reviewer (employee lookup)
- Approver (employee lookup)
- Tags (optional free text)

Actions:

  - Save as Draft
  - Submit for Approval

Screen: My Assets Screen

Fields / Sections:

- Table: Asset Name | Type | Version | Status | Compliance Score | Last Updated
- Filter: by type, status, compliance status
- My Owned | My Contributed tabs

Actions:

  - Open Asset
  - Register New Asset
  - View Compliance Status

Screen: Asset Detail Screen

Fields / Sections:

- Asset header: Name | Type | Version | Status | Compliance Score
- Ownership panel: Creator | Contributors | Reviewer | Approver | Current Owner
- Documentation section: required docs checklist with upload/compliance status per item

- Version History table: version | date | changed by | change summary | major/minor
- Activity log: all actions on this asset with timestamp and user

Actions:

  - Upload Documentation
  - Submit New Version
  - Transfer Ownership (HR Admin only)
  - Archive Asset (HR Admin only)

Screen: Exit Handover Checklist Screen

Fields / Sections:

- Exiting employee name and exit date
- Table: Asset Name | Type | Creator/Contributor | Current Owner | Handover Recipient (assignable)
- Unassigned assets count badge
- Confirmation button (disabled until all assets assigned)

Actions:

  - Assign Recipient per asset
  - Bulk assign to one recipient
  - Confirm Handover

11.4 Business Rules & Validations

- **BR-01:** Asset Type is mandatory and must be selected from Asset Type Master. Free-text asset types are not permitted.
- **BR-02:** An asset must have exactly one Creator — the employee who originally created it. The Creator field is set on registration and is immutable.
- **BR-03:** An asset can have zero or more Contributors. Contributors are listed separately from the Creator.
- **BR-04:** Reviewer and Approver must be different employees and cannot be the same as the Creator.
- **BR-05:** Asset version numbering follows Major.Minor format. Major version is incremented for significant revisions (employee selects). Minor for patches. Version numbers are auto-incremented by the system.
- **BR-06:** Every version is stored — no version is deleted. Previous versions are archived (read-only).
- **BR-07:** When an employee exits, all assets where they are the Current Owner must have a handover recipient assigned before exit confirmation is allowed.
- **BR-08:** After exit handover, the Creator, Contributors, and historical version records remain attributed to the original (now exited) employee. Only the Current Owner changes.
- **BR-09:** Asset status lifecycle: Draft → Pending Approval → Active → Archived / Deprecated. Archived assets are read-only.

- **BR-10:** Asset records are immutable once Active — ownership metadata changes (handover) are recorded as events in the activity log, not as edits to the original fields.

11.5 Function Signatures

Function	Input	Process Summary	Output
registerAsset(empId, assetData)	Employee ID, asset data	Validates type, creates asset v1.0, attaches doc checklist, notifies reviewer/approver	AssetRecord | ValidationError
approveAsset(assetId, approverId)	Asset ID, Approver ID	Updates status to Active, activates compliance tracking, records approval	ApprovedAsset
submitNewVersion(assetId, empId, versionData)	Asset ID, Employee ID, version type, changes	Archives current version, creates new version, updates activity log	NewAssetVersion
generateExitHandoverList(empId)	Employee ID	Returns all assets where employee is Creator or Owner with handover status	HandoverChecklist[]
transferOwnership(assetId, newOwnerId, hrAdminId, reason)	Asset ID, new Owner ID, HR Admin ID, reason	Updates Current Owner, records event in activity log, does not alter Creator or Contributors	TransferEvent
searchAssets(filters)	Filter object: type, status, owner, department, compliance	Returns paginated asset list matching filters	PaginatedAssetList

12. Asset & Documentation Management

12.1 Overview

This module enforces documentation requirements for every registered asset. Required documentation types per asset category are defined in Doc Requirement Config. Compliance is automatically calculated and monitored. Non-compliance actions are configurable.

Actors / Roles	Description
Employee	Uploads documentation for owned and contributed assets
Manager	Reviews documentation compliance for team assets
HR Admin	Configures documentation requirements and compliance thresholds; views non-compliance reports

12.2 User Flows & Workflows

Flow: Employee Uploads Documentation

1. Employee opens an asset from My Assets.
2. Views the Documentation section — a checklist of required items for the asset type.
3. For each required item, employee uploads the relevant file or enters the required text/link.
4. System marks each item as: Submitted / Missing.
5. System recalculates Documentation Compliance Score = (Submitted / Required) × 100.
6. If score rises above the threshold, the non-compliance flag is cleared.
7. If score falls below threshold, flag remains and notifications are sent per Non-Compliance Action Config.

Flow: System Monitors Compliance

1. A daily compliance job runs across all Active assets.
2. For each asset, compliance score is recalculated.
3. Assets below the threshold defined in Compliance Threshold Config are flagged.
4. Non-Compliance Action Config determines the consequence: notification only / affect efficiency score / block asset / escalate to manager.
5. HR Admin receives a daily Non-Compliance Report.

Flow: HR Admin Configures Documentation Requirements

1. HR Admin opens Admin Dashboard → Documentation Requirements.
2. Selects an Asset Type.
3. Views current required documentation items for that type.
4. Adds, removes, or reorders items.

5. Changes take effect for new assets immediately. Existing assets are flagged for re-compliance review.

12.3 Screen Descriptions

Screen: Documentation Section (within Asset Detail Screen)

Fields / Sections:

- Compliance Score badge (% with color: green ≥ threshold, amber near, red below)
- Required docs checklist: item name | status (Submitted/Missing) | uploaded file link | upload date
- Non-compliant banner if below threshold

Actions:

  - Upload file for each item
  - Replace existing upload
  - Mark item as Not Applicable (with reason, subject to manager approval)

Screen: Non-Compliance Report Screen (HR Admin)

Fields / Sections:

- Summary: total active assets | compliant | non-compliant | compliance rate %
- Table: Asset Name | Type | Owner | Compliance Score | Missing Items count | Days Non-Compliant | Action Applied
- Filter: by asset type, department, compliance threshold breach level

Actions:

  - Open Asset
  - Notify Owner
  - Export Report

Screen: Documentation Requirement Config Screen (Admin)

Fields / Sections:

- Asset Type selector
- Current required items list: item name | mandatory/optional flag | description
- Add item form
- Effective date

Actions:

  - Add Item
  - Remove Item (with confirmation and impact warning)
  - Save Config Version

12.4 Business Rules & Validations

- **BR-01:** Documentation compliance is calculated as: (number of submitted required items) / (total required items for asset type) × 100. Optional items do not count toward compliance calculation.
- **BR-02:** Compliance Threshold is read from Compliance Threshold Config for the asset's type. If no type-specific threshold is set, the system-level default threshold applies.
- **BR-03:** A documentation item marked Not Applicable (N/A) counts as Submitted toward the compliance score, provided a manager has approved the N/A designation.
- **BR-04:** Changes to Doc Requirement Config that add new required items cause all existing assets of that type to have their compliance score recalculated immediately.
- **BR-05:** Non-compliance consequences are applied per Non-Compliance Action Config — the system does not hardcode any consequence.
- **BR-06:** File uploads are stored with: uploader, upload timestamp, file name, file size, and file hash (for integrity verification).
- **BR-07:** Duplicate file uploads for the same documentation item replace the previous upload. The previous file is archived, not deleted.
- **BR-08:** Assets in Draft or Pending Approval status are excluded from compliance monitoring — monitoring begins only after status is Active.

12.5 Function Signatures

Function	Input	Process Summary	Output
uploadDocumentation(assetId, docItemId, empId, fileRef)	Asset ID, doc item ID, Employee ID, file reference	Stores file, marks item as Submitted, recalculates compliance score, triggers action if below threshold	ComplianceUpdate
calculateComplianceScore(assetId)	Asset ID	Reads required items from Doc Requirement Config, counts submitted vs. required, returns score	ComplianceScore {score, submitted, required, missing[]}
flagNonCompliantAssets()	None (system job)	Scans all active assets, flags non-compliant ones, applies actions per config, generates report	NonComplianceReport
approveNotApplicable(assetId, docItemId, managerId, reason)	Asset ID, doc item ID, Manager ID, reason	Marks item as N/A (counts as submitted), records approval	N/AApproval
getDocRequirementConfig(assetTypeId)	Asset Type ID	Returns active documentation requirements for the asset type	DocRequirementList[]

13. Contribution Tracking

13.1 Overview

Contribution Tracking links employee work (tasks, time, assets) into a unified contribution record. It provides a traceable history of every employee's contributions to organizational assets and projects, which feeds into the Reward Engine.

Actors / Roles	Description
System	Auto-records contributions when tasks are accepted and assets are registered
HR Admin	Views and reports on contribution data; links tasks to assets
Manager	Views team contribution summaries
Employee	Views own contribution history

13.2 User Flows & Workflows

Flow: Auto-Record Contribution on Task Acceptance

1. Manager accepts a task.
2. System automatically creates a Contribution Record: employee ID, task ID, acceptance date, quality rating, time logged.
3. If the task is linked to an asset, the contribution is also linked to the asset record.
4. Contribution is visible in the employee's Contribution History.

Flow: Link Task to Asset

1. HR Admin or Employee opens a task and selects 'Link to Asset'.
2. Selects the target asset from the Asset Registry.
3. System links the task's contribution record to the asset.
4. The asset's contribution log is updated.

Flow: View Contribution Report

1. HR Admin opens Reports → Contribution Report.
2. Selects employee and date range.
3. System returns: list of all accepted tasks, assets owned, assets contributed to, quality ratings, efficiency scores, and linked rewards.
4. Report is exportable.

13.3 Screen Descriptions

Screen: My Contributions Screen — Employee

Fields / Sections:

- Timeline of contributions: task name | asset linked | date | quality rating | time logged
- Summary: total tasks accepted | assets created | assets contributed | avg quality rating
- Period filter

Actions:

  - View Task Detail
  - View Asset Detail
  - Export

Screen: Contribution Report — HR Admin

Fields / Sections:

- Employee selector
- Period selector
- Contribution table: date | type (task/asset) | name | quality | time | linked asset | reward linked
- Totals at the bottom

Actions:

  - Export PDF / CSV
  - View Reward History for this employee

13.4 Business Rules & Validations

- **BR-01:** Contribution records are created automatically by the system — they do not require manual entry.
- **BR-02:** A contribution record is created only when a task is Accepted. Rejected or in-progress tasks do not generate contribution records.
- **BR-03:** Contribution records are immutable once created. They cannot be edited or deleted.
- **BR-04:** An employee's contribution to an asset is recorded when they are listed as Creator, Contributor, Reviewer, or Approver on that asset.
- **BR-05:** Contribution data is used by the Reward Engine to determine reward eligibility and split percentages across contributors.

13.5 Function Signatures

Function	Input	Process Summary	Output
recordContribution(empId, taskId, assetId?)	Employee ID, Task ID, optional Asset ID	Creates contribution record on task acceptance, links to asset if provided	ContributionRecord
linkTaskToAsset(taskId, assetId, userId)	Task ID, Asset ID, acting user ID	Links an accepted task's contribution to an asset record	LinkConfirmation

Function	Input	Process Summary	Output
getContributionHistory(empId, period)	Employee ID, period	Returns all contribution records for the employee in the period	ContributionHistory[]
getAssetContributions(assetId)	Asset ID	Returns all contributor records linked to an asset	AssetContributions[]

14. Reward & Incentive Management

14.1 Overview

The Reward Engine calculates and manages long-term incentives tied to the business value of employee-created assets. All reward types, calculation formulas, contribution splits, payment schedules, and caps are metadata-driven and configurable by HR and Finance.

Actors / Roles	Description
HR Admin	Nominates assets for rewards, manages reward program configuration
Finance Admin	Configures reward formulas and caps, approves reward calculations, manages payment schedules
Employee	Views own reward eligibility and history
System	Calculates reward amounts, splits across contributors, schedules payments

14.2 User Flows & Workflows

Flow: Nominate Asset for Reward

1. HR Admin opens Reward Management → New Nomination.
2. Selects the asset from the Asset Registry.
3. Selects the reward basis (from Reward Basis Master: revenue generated / cost savings / IP value / etc.).
4. Enters the measured value (e.g., revenue amount, savings amount).
5. Selects reward type (from Reward Type Master: annual bonus / royalty / innovation incentive / etc.).
6. Submits nomination for Finance Admin review.

Flow: Finance Admin Reviews and Calculates Reward

1. Finance Admin opens Reward Nominations → Pending Review.
2. Reviews the nomination: asset, basis, measured value, reward type.
3. System shows the calculation preview using Reward Formula Config for the selected reward type.
4. Finance Admin reviews the split across Creator and Contributors per Reward Contribution Config.
5. Finance Admin checks against Reward Cap Config — if any individual amount exceeds the cap, the system flags it.
6. Finance Admin approves. System generates reward records per contributor.
7. Payment is scheduled per Reward Period Config.

Flow: Employee Views Reward History

1. Employee navigates to My Rewards.

2. Views: all reward nominations linked to their assets, reward type, calculated amount, payment status, payment date.
3. Views beneficiary designation status.

14.3 Screen Descriptions

Screen: Reward Nomination Form

Fields / Sections:

- Asset selector (from Asset Registry)
- Reward Basis selector (from Reward Basis Master)
- Measured value input (numeric, with currency)
- Reward Type selector (from Reward Type Master)
- Supporting notes

Actions:

  - Submit for Finance Review
  - Save Draft

Screen: Reward Calculation Preview Screen (Finance Admin)

Fields / Sections:

- Asset detail: name | type | creator | contributors
- Reward basis: metric | measured value
- Formula applied (from Reward Formula Config): formula description + inputs
- Calculated reward total
- Contributor split table: name | role | split % | amount | cap check status
- Payment schedule preview per Reward Period Config

Actions:

  - Approve and Generate Rewards
  - Reject with Reason
  - Edit Measured Value and Recalculate

Screen: My Rewards Screen — Employee

Fields / Sections:

- Table: Asset Name | Reward Type | Basis | My Share Amount | Payment Status | Payment Date | Beneficiary
- Filter by status and year
- Total rewards earned this year

Actions:

  - View Asset Detail

- View Beneficiary Assignment

14.4 Business Rules & Validations

- **BR-01:** Reward nominations must specify a reward basis from Reward Basis Master. Free-text reward basis is not permitted.
- **BR-02:** Reward calculation uses the formula version that was active at the time of nomination — not the current latest formula.
- **BR-03:** Contributor split percentages must sum to 100% across Creator and all Contributors. The split method is defined in Reward Contribution Config.
- **BR-04:** Reward Cap Config is applied per contributor per reward type per period. If a calculated amount exceeds the cap, the excess is flagged for Finance Admin decision — it is not automatically redistributed.
- **BR-05:** Approved reward records are locked. Finance Admin can only make corrections through a mandatory audit log entry.
- **BR-06:** If an employee has exited the organization but is still a Creator or Contributor on an active asset, they remain eligible for rewards unless organizational policy prohibits it. This is configured in Reward Formula Config.
- **BR-07:** Payment schedule is set per Reward Period Config — the system generates a payment schedule entry but does not execute actual payments (payroll disbursement is out of scope).
- **BR-08:** Reward records are retained permanently. No reward record is deleted.

14.5 Function Signatures

Function	Input	Process Summary	Output
nominateAssetForReward(hrAdminId, nominationData)	HR Admin ID, nomination data	Validates asset, basis, and reward type; creates nomination; notifies Finance Admin	NominationRecord | ValidationError
calculateReward(nominationId)	Nomination ID	Reads formula config, computes reward total, splits across contributors per Reward Contribution Config, checks caps	RewardCalculation {total, splits[], capFlags[]}
approveReward(nominationId, financeAdminId)	Nomination ID, Finance Admin ID	Locks calculation, generates contributor reward records, schedules payment	RewardRecords[]
getEmployeeRewards(empId)	Employee ID	Returns all reward records for the employee across all assets	RewardHistory[]
saveRewardFormulaConfig(rewardTypeId, formula, effectiveDate)	Reward Type ID, formula definition, effective date	Creates versioned formula config	ConfigVersion

15. Beneficiary Management

15.1 Overview

Employees may nominate one or more beneficiaries to receive their eligible long-term rewards. Beneficiary types, required documents, and jurisdiction rules are all metadata-driven. Allocation percentages must sum to 100%. HR verifies documentation before beneficiaries are activated.

Actors / Roles	Description
Employee	Nominates beneficiaries, uploads required documents, manages allocation
HR Admin	Verifies beneficiary documents, activates beneficiaries, manages configuration

15.2 User Flows & Workflows

Flow: Employee Nominates Beneficiary

1. Employee navigates to My Profile → Beneficiaries → Add Beneficiary.
2. Selects beneficiary type (from Beneficiary Type Master: spouse / child / NGO / trust / etc.).
3. Enters beneficiary name, contact details, and relationship.
4. Enters allocation percentage for this beneficiary.
5. System shows required documents for this beneficiary type (from Beneficiary Doc Config).
6. Employee uploads required documents.
7. Saves. Beneficiary status: Pending Verification.
8. HR Admin is notified to verify.

Flow: HR Admin Verifies Beneficiary

1. HR Admin opens Beneficiary Verification Queue.
2. Reviews submitted documents against Beneficiary Doc Config requirements.
3. If all documents are valid: marks beneficiary as Verified. Status changes to Active.
4. If documents are insufficient: marks as Rejected with a reason. Employee is notified and can resubmit.
5. Active beneficiaries are eligible to receive reward payments.

Flow: Employee Updates Beneficiary Allocation

1. Employee opens Beneficiaries tab.
2. Adjusts allocation percentages across their active beneficiaries.
3. System validates that total allocation = 100%.
4. Saves. Allocation update takes effect for future reward payments.

15.3 Screen Descriptions

Screen: Beneficiary List Screen — Employee

Fields / Sections:

- Table: Beneficiary Name | Type | Relationship | Allocation % | Status | Verification Date
- Total allocation % counter (must equal 100% for full coverage)
- Add Beneficiary button

Actions:

  - Add Beneficiary
  - Edit Allocation
  - Remove Beneficiary (with confirmation)
  - View Required Documents

Screen: Beneficiary Detail / Add Form

Fields / Sections:

- Beneficiary Type (dropdown from Beneficiary Type Master)
- Name (text, mandatory)
- Relationship to Employee (text)
- Contact Details
- Allocation % (numeric)
- Required Documents checklist (from Beneficiary Doc Config — per type)
- Document uploads per required item

Actions:

  - Save
  - Submit for Verification
  - Cancel

Screen: Beneficiary Verification Queue — HR Admin

Fields / Sections:

- Table: Employee Name | Beneficiary Name | Type | Submitted Date | Documents Submitted / Required
- Filter: by type, status
- Click row to open detail and verify documents

Actions:

  - Verify (Activate)
  - Reject with Reason
  - Request Additional Documents

15.4 Business Rules & Validations

- **BR-01:** Beneficiary type must be selected from Beneficiary Type Master. Free-text beneficiary types are not permitted.
- **BR-02:** If an employee has multiple beneficiaries, allocation percentages must sum to exactly 100% before the beneficiary setup is considered complete. Partial allocations (summing to less than 100%) are allowed in draft state but flagged as incomplete.
- **BR-03:** Required documents per beneficiary type are defined in Beneficiary Doc Config. A beneficiary cannot be activated until all required documents are uploaded and verified by HR Admin.
- **BR-04:** A beneficiary can be Removed by the employee at any time. If removal causes total allocation to drop below 100%, the employee is prompted to reallocate.
- **BR-05:** If no active beneficiary is designated at the time a reward becomes payable, the reward is flagged to HR Admin for manual resolution per organizational policy.
- **BR-06:** Beneficiary records and all uploaded documents are retained permanently. Removed beneficiaries are archived, not deleted.
- **BR-07:** Jurisdiction rules from Jurisdiction Config may impose additional requirements (e.g., legal notarization for certain beneficiary types in certain regions). The system displays jurisdiction-specific requirements based on the employee's location.
- **BR-08:** Allocation changes do not affect previously scheduled reward payments. They apply only to future payment schedules.

15.5 Function Signatures

Function	Input	Process Summary	Output
nominateBeneficiary(empId, beneficiaryData)	Employee ID, beneficiary data	Validates type and documents, creates pending beneficiary record, notifies HR Admin	BeneficiaryRecord | ValidationError
verifyBeneficiary(beneficiaryId, hrAdminId, decision, reason?)	Beneficiary ID, HR Admin ID, decision, optional rejection reason	Updates beneficiary status, notifies employee	UpdatedBeneficiary
updateAllocation(empId, allocations)	Employee ID, array of {beneficiaryId, allocationPct}	Validates total = 100%, updates allocations for future payments	UpdatedAllocations | ValidationError
getBeneficiaries(empId)	Employee ID	Returns all beneficiaries (active, pending, archived) for the employee	BeneficiaryList[]
getRequiredDocuments(beneficiaryTypeId, jurisdictionId)	Beneficiary Type ID, Jurisdiction ID	Returns required document checklist for the type and jurisdiction	DocRequirementList[]

16. HR & Finance Admin Dashboard

16.1 Overview

The Admin Dashboard is the central configuration and oversight hub. HR and Finance Admins manage all configuration entities, monitor system health, review pending actions, and access cross-module reports. Every configuration change is versioned and logged.

Actors / Roles	Description
HR Admin	Manages all HR configuration entities, employee data, reward nominations, compliance monitoring
Finance Admin	Manages salary bands, reward formulas, approves salary and reward calculations
System Admin	Manages user accounts, roles, permissions, audit logs, system-level configuration

16.2 User Flows & Workflows

Flow: Admin Updates a Configuration Entity

1. Admin navigates to Admin Dashboard → Configuration Management → selects the entity (e.g., BPV Weight Config).
2. Views the currently active configuration version.
3. Clicks Edit / Add New Version.
4. Makes changes to values.
5. Enters effective date (must be today or future).
6. Enters a mandatory change reason (minimum 20 characters).
7. System validates the config (e.g., weights sum to 100%).
8. On save: old config record is closed (effective_to = day before new effective_from). New version is inserted.
9. Admin receives confirmation. System-wide cache is invalidated and refreshed.

Flow: Admin Manages Pending Actions

1. Admin Dashboard home shows an Inbox of pending actions requiring attention.
2. Examples: BPV recalculation queue, salary period approvals, reward nominations, beneficiary verifications, non-compliance alerts, exit handovers.
3. Admin clicks an item → navigated to the relevant module for action.

16.3 Screen Descriptions

Screen: Admin Dashboard Home

Fields / Sections:

- Inbox panel: pending actions count by category (salary approvals, reward reviews, non-compliance, verifications)
- System health indicators: last BPV run | last efficiency run | config cache status
- Quick navigation: all configuration entity categories

Actions:

  - Navigate to any config entity
  - Open pending action items
  - View audit log

Screen: Configuration Management Hub

Fields / Sections:

- Categorized list of all configuration entities (BPV, Time, Salary, Efficiency, Ownership, Docs, Rewards, Beneficiary, System)
- For each entity: current version effective date | last changed by | last changed date
- Click entity → open its configuration screen

Actions:

  - Open any config entity
  - View version history for any entity
  - Export config snapshot

Screen: Audit Log Screen

Fields / Sections:

- Table: timestamp | user | module | entity | action | old value | new value | reason
- Filter: by user, module, date range, action type
- Search by entity name

Actions:

  - Export audit log
  - View full change detail

16.4 Business Rules & Validations

- **BR-01:** Every configuration change must include a change reason of at least 20 characters. Changes without a reason are rejected.
- **BR-02:** Effective dates for configuration changes cannot be in the past. Backdated configurations are not permitted.
- **BR-03:** When a configuration entity is updated, the system invalidates the relevant cache and rebuilds it — ensuring all subsequent calculations use the new configuration.

- **BR-04:** HR Admin and Finance Admin have role-specific access to configuration entities. HR Admin cannot edit Finance-only entities (salary bands, reward formulas) and vice versa.
- **BR-05:** System Admin can view all configuration entities but cannot edit HR or Finance domain configurations without explicit permission.
- **BR-06:** The audit log is append-only. No audit record can be edited or deleted. Audit logs are retained per Audit Log Config retention period.
- **BR-07:** All changes to configuration entities trigger a system notification to the System Admin.

16.5 Function Signatures

Function	Input	Process Summary	Output
updateConfig(entityType, configData, effectiveDate, reason, adminId)	Entity type, new config data, effective date, reason, admin ID	Validates config, closes old version, creates new version, invalidates cache, logs audit entry	ConfigVersion | ValidationError
getConfigHistory(entityType, entityId?)	Entity type, optional entity ID	Returns full version history for the configuration entity	ConfigVersionHistory[]
getPendingActions(adminId)	Admin ID	Returns all pending actions requiring the admin's attention, grouped by category	PendingActionsList[]
getAuditLog(filters)	Filter object: user, module, entity, date range, action	Returns filtered audit log entries	AuditLogList[]
invalidateConfigCache(entityType)	Entity type	Clears the cache for the specified configuration entity and triggers a rebuild	CacheRefreshConfirmation

17. Reports & Analytics

17.1 Overview

The Reports & Analytics module provides cross-module reporting for HR, Finance, and Management. All reports respect role-based data access. Reports can be run on-demand, scheduled, and exported in multiple formats.

Actors / Roles	Description
HR Admin	Runs all HR reports: BPV summary, efficiency, compliance, contribution, workforce analytics
Finance Admin	Runs salary, reward, and financial analytics reports
Manager	Runs team-level performance, efficiency, and task reports (own team only)
Employee	Views own performance summary and contribution report
System Admin	Runs system audit and configuration history reports

17.2 User Flows & Workflows

Flow: Run On-Demand Report

1. Admin or Manager navigates to Reports → Report Library.
2. Selects report type.
3. Applies filters (date range, department, designation, employee, asset type, etc.).
4. Clicks Run Report.
5. System generates report. Results displayed in tabular and chart form.
6. User downloads in PDF or CSV format.

Flow: Schedule a Report

1. User selects a report and clicks Schedule.
2. Sets frequency: daily / weekly / monthly.
3. Sets recipients (email addresses).
4. Report runs automatically and is emailed to recipients.

17.3 Screen Descriptions

Screen: Report Library Screen

Fields / Sections:

- Grid of available reports grouped by category: Workforce, BPV & Salary, Performance, Asset & Compliance, Reward, Audit
- Each report card: report name | description | last run | schedule status

Actions:

- Run Report
- Schedule Report
- View Scheduled Reports

Screen: Report Viewer Screen

Fields / Sections:

- Report title and run parameters (date range, filters applied)
- Summary KPIs at the top
- Data table with sortable columns
- Charts where applicable (bar, line, pie)
- Export buttons: PDF | CSV | Excel

Actions:

- Download
- Share Link (authenticated)
- Schedule this report
- Rerun with different filters

17.4 Business Rules & Validations

- **BR-01:** Data access in reports is strictly role-controlled. Managers can only see data for employees in their direct team. HR Admin and Finance Admin see data per their domain access. Employees see only their own data.
- **BR-02:** Reports cannot expose salary data to managers. Salary fields are hidden from manager-role reports.
- **BR-03:** Scheduled reports are sent only to authorized recipients. The system validates recipient email addresses against the user directory.
- **BR-04:** All report runs are logged in the audit trail: who ran it, when, with what filters, and how many records were returned.
- **BR-05:** Report data reflects the state of the system at the time the report is run — it is not cached from a previous run.

17.5 Function Signatures

Function	Input	Process Summary	Output
generateReport(reportType, filters, userId)	Report type, filter parameters, requesting user ID	Validates role access, queries relevant modules, compiles report data	ReportResult {data[], summary{}, charts[]}
exportReport(reportId, format)	Report run ID, format (pdf/csv/xlsx)	Generates file in requested format	FileDownloadUrl

Function	Input	Process Summary	Output
scheduleReport(reportType, filters, frequency, recipients, userId)	Report config, schedule settings, recipient list	Creates scheduled job, validates recipients	ScheduleConfirmation
getScheduledReports(userId)	User ID	Returns all scheduled reports for the user	ScheduledReportList[]

18. Document Control

Field	Value
Document Title	HRMS — Functional Design Document
Version	v1.0
Status	Draft — For Review
Date	July 2026
Classification	Confidential
Modules Covered	15 Functional Modules + Admin Dashboard + Reports (17 sections)
Companion Document	HRMS Solution Design Document v2.0 (Metadata-Driven Architecture)
Intended Audience	UI/UX Designers, Frontend & Backend Developers, QA Engineers, Project Managers

All business rules described in this document are enforced at runtime using the configuration entities defined in the Solution Design Document. Any rule labeled as configurable will not be hardcoded in the application.